FairLens - Bias Audit Report

Fairness, Performance and Explainability Analysis

1. Dataset Summary

Property	Value
Dataset Rows	32537
Dataset Columns	15
Target Column	Unknown
Sensitive Features	sex, race

2. Model Information

Property	Value
Model	Random Forest Classifier
Task Type	classification
Sensitive Attributes	sex, race

3. Model Performance

Metric	Before Mitigation	After Mitigation
Accuracy	0.8423478795328826	0.8408113091579594
Precision	0.842560709065893	0.8419713622159438
Recall	0.8423478795328826	0.8408113091579594
F1 Score	0.8220187270572272	0.8192812810046232

4. Fairness Evaluation

Stage	Sensitive Attribute	Metric	Value
original	sex	Demographic Parity Difference	0.0000
original	sex	Disparate Impact	1.0000
original	sex	Equal Opportunity Difference	0.0000
original	sex	Group Accuracy	Male: 0.8053 Female: 0.9162
original	sex	Equalized Odds	0.0000
original	race	Demographic Parity Difference	0.0000
original	race	Disparate Impact	1.0000
original	race	Equal Opportunity Difference	0.0000
original	race	Group Accuracy	White: 0.8346 Black: 0.9003 Asian-Pac-Islander: 0.8342 Amer-Indian-Eskimo: 0.9057 Other: 0.9535

Stage	Sensitive Attribute	Metric	Value
original	race	Equalized Odds	0.0000
mitigated	sex	Demographic Parity Difference	0.0000
mitigated	sex	Disparate Impact	1.0000
mitigated	sex	Equal Opportunity Difference	0.0000
mitigated	sex	Group Accuracy	Male: 0.8028 Female: 0.9167
mitigated	sex	Equalized Odds	0.0000
mitigated	race	Demographic Parity Difference	0.0000
mitigated	race	Disparate Impact	1.0000
mitigated	race	Equal Opportunity Difference	0.0000
mitigated	race	Group Accuracy	White: 0.8321 Black: 0.9019 Asian-Pac-Islander: 0.8492 Amer-Indian-Eskimo: 0.9057 Other: 0.9535
mitigated	race	Equalized Odds	0.0000

5. Bias Assessment

Overall Assessment: No significant bias detected

6. Mitigation Analysis

Recommended Mitigation

Sensitive Attribute	Bias Detected	Recommended Strategy	Reason
sex	No	No mitigation required	No significant bias was detected.
race	No	No mitigation required	No significant bias was detected.

Applied Mitigation

Reweighing

7. Before vs After Comparison

Performance

Metric	Before Mitigation	After Mitigation
Accuracy	0.8423478795328826	0.8408113091579594
Precision	0.842560709065893	0.8419713622159438
Recall	0.8423478795328826	0.8408113091579594
F1 Score	0.8220187270572272	0.8192812810046232

Fairness

Stage	Sensitive Attribute	Metric	Value
original	sex	Demographic Parity Difference	0.0000
original	sex	Disparate Impact	1.0000

Stage	Sensitive Attribute	Metric	Value
original	sex	Equal Opportunity Difference	0.0000
original	sex	Group Accuracy	Male: 0.8053
original	sex	Group Accuracy	Female: 0.9162
original	sex	Equalized Odds	0.0000
original	race	Demographic Parity Difference	0.0000
original	race	Disparate Impact	1.0000
original	race	Equal Opportunity Difference	0.0000
original	race	Group Accuracy	White: 0.8346
original	race	Group Accuracy	Black: 0.9003
original	race	Group Accuracy	Asian-Pac-Islander: 0.8342
original	race	Group Accuracy	Amer-Indian-Eskimo: 0.9057
original	race	Group Accuracy	Other: 0.9535
original	race	Equalized Odds	0.0000
mitigated	sex	Demographic Parity Difference	0.0000
mitigated	sex	Disparate Impact	1.0000
mitigated	sex	Equal Opportunity Difference	0.0000
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mitigated	race	Group Accuracy	White: 0.8321
mitigated	race	Group Accuracy	Black: 0.9019
mitigated	race	Group Accuracy	Asian-Pac-Islander: 0.8492
mitigated	race	Group Accuracy	Amer-Indian-Eskimo: 0.9057
mitigated	race	Group Accuracy	Other: 0.9535
mitigated	race	Equalized Odds	0.0000

8. Explainability

SHAP feature importance is not available for this model.

9. Recommendations

Sensitive Attribute	Bias Detected	Recommended Strategy	Reason
sex	No	None	No significant bias was detected.
race	No	None	No significant bias was detected.

10. Final Conclusion

A mitigation strategy was applied. Review the before and after fairness metrics together with model performance to determine whether the mitigated model provides an appropriate fairness and performance trade-off.